

HDOC DAY – 31

ALGORITHM:

1. Start.
2. Input the number of rows and columns.
3. Input all elements of the matrix.
4. Display the menu:
 - Sum of all elements
 - Sum of each row and store in an array
 - Transpose of matrix
 - Exit
5. Read the user's choice.
6. Use switch-case:
 - **Choice 1:** Add all matrix elements and display the total.
 - **Choice 2:** Find the sum of each row, store each sum in a 1D array, and display the array.
 - **Choice 3:** Display the transpose by interchanging rows and columns.
 - **Choice 4:** Exit.
 - Otherwise, display an invalid-choice message.
7. Repeat the menu until Choice 4 is selected.
8. Stop.

C PROGRAM:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int rows, cols;
```

```
    int i, j, choice;
```

```
int sum;
```

```
printf("Enter the number of rows and columns: ");
```

```
scanf("%d %d", &rows, &cols);
```

```
int matrix[rows][cols];
```

```
int rowSum[rows];
```

```
printf("\nEnter the matrix elements:\n");
```

```
for (i = 0; i < rows; i++)
```

```
{
```

```
    for (j = 0; j < cols; j++)
```

```
    {
```

```
        scanf("%d", &matrix[i][j]);
```

```
    }
```

```
}
```

```
do
```

```
{
```

```
    printf("\nMenu\n");
```

```
    printf("1. Find sum of all elements\n");
```

```
    printf("2. Find sum of each row\n");
```

```
    printf("3. Find transpose of matrix\n");
```

```
    printf("4. Exit\n");
```

```
    printf("\nEnter your choice: ");
```

```
    scanf("%d", &choice);
```

```
switch (choice)
{
    case 1:
        sum = 0;

        for (i = 0; i < rows; i++)
        {
            for (j = 0; j < cols; j++)
            {
                sum = sum + matrix[i][j];
            }
        }

        printf("Sum of all elements = %d\n", sum);
        break;

    case 2:
        printf("\nRow sums:\n");

        for (i = 0; i < rows; i++)
        {
            rowSum[i] = 0;

            for (j = 0; j < cols; j++)
            {
                rowSum[i] = rowSum[i] + matrix[i][j];
            }
        }
    }
}
```

```
    printf("Sum of row %d = %d\n", i + 1, rowSum[i]);  
}
```

```
printf("\nThe row-sum array contains: [");
```

```
for (i = 0; i < rows; i++)  
{  
    printf("%d", rowSum[i]);
```

```
    if (i < rows - 1)  
    {  
        printf(", ");  
    }  
}
```

```
printf("]\n");
```

```
break;
```

case 3:

```
printf("\nTranspose of the matrix:\n");
```

```
for (j = 0; j < cols; j++)  
{  
    for (i = 0; i < rows; i++)  
    {  
        printf("%d\t", matrix[i][j]);  
    }  
}
```

```

        printf("\n");
    }
    break;

case 4:
    printf("Exiting the program.\n");
    break;

default:
    printf("Invalid choice! Please enter a choice from 1 to 4.\n");
}

} while (choice != 4);

return 0;
}

```

TEST CASE TABLE:

TEST CASE	MATRIX	CHOICE	EXPECTED RESULT
1	1 2 3 / 4 5 6	1	Sum = 21
2	1 2 3 / 4 5 6	2	Row-sum array = [6, 15]
3	1 2 3 / 4 5 6	3	Transpose = 1 4 / 2 5 / 3 6
4	Any valid matrix	4	Program exits
5	Any valid matrix	5	Invalid choice!